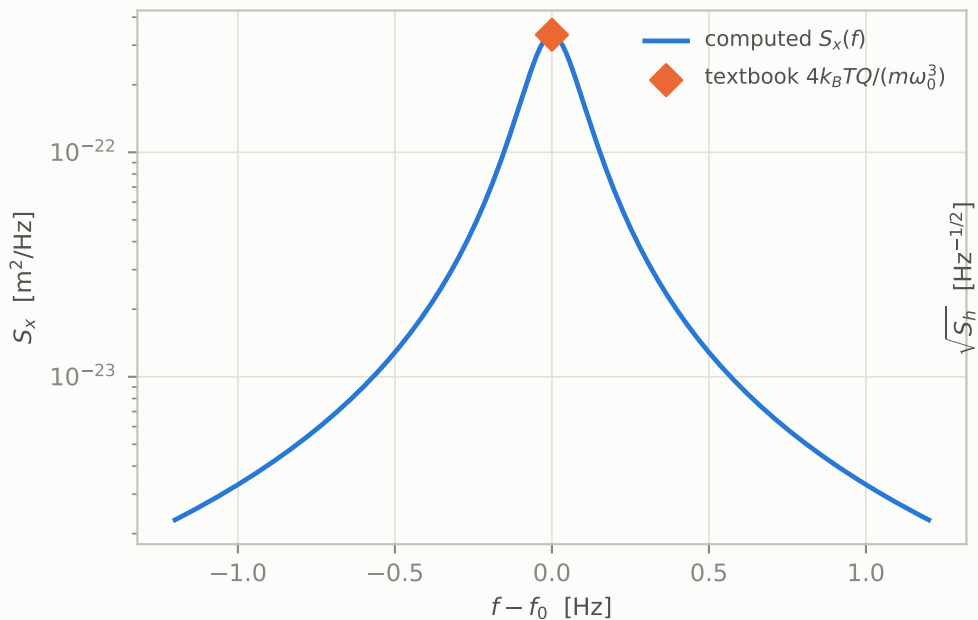
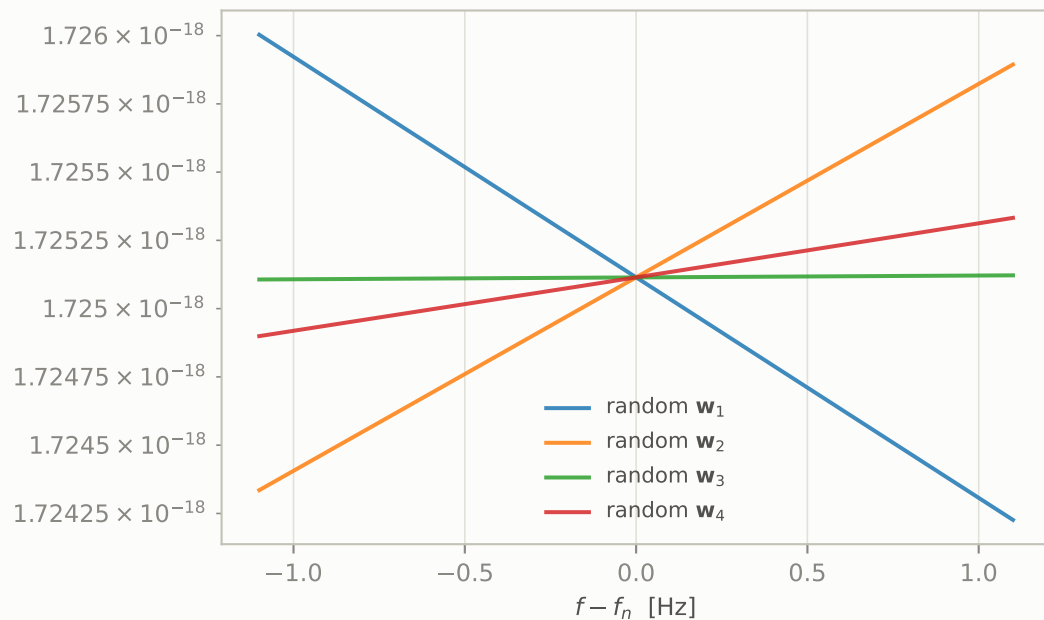


physical.py validation suite — all tests passing

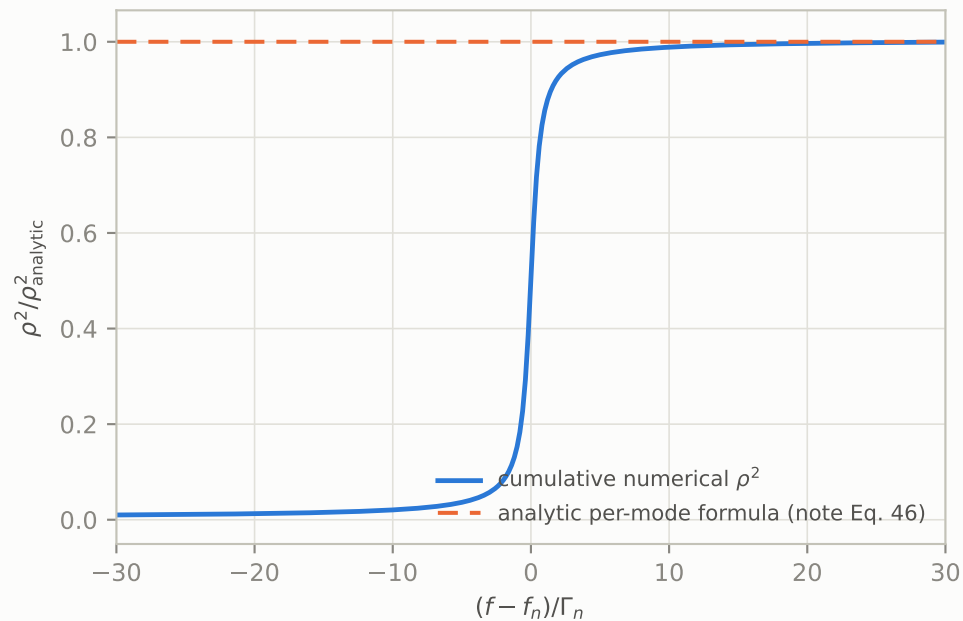
1) FDT thermal peak — agreement to 10^{-6}



2) thermal $\sqrt{S_h}$ on resonance is weight-independent (spread 7e-09)



3) matched-filter SNR: numerical/analytic = 1.0075



4) benchmark $\sqrt{S_h}$ — gap = unconfirmed parameters

